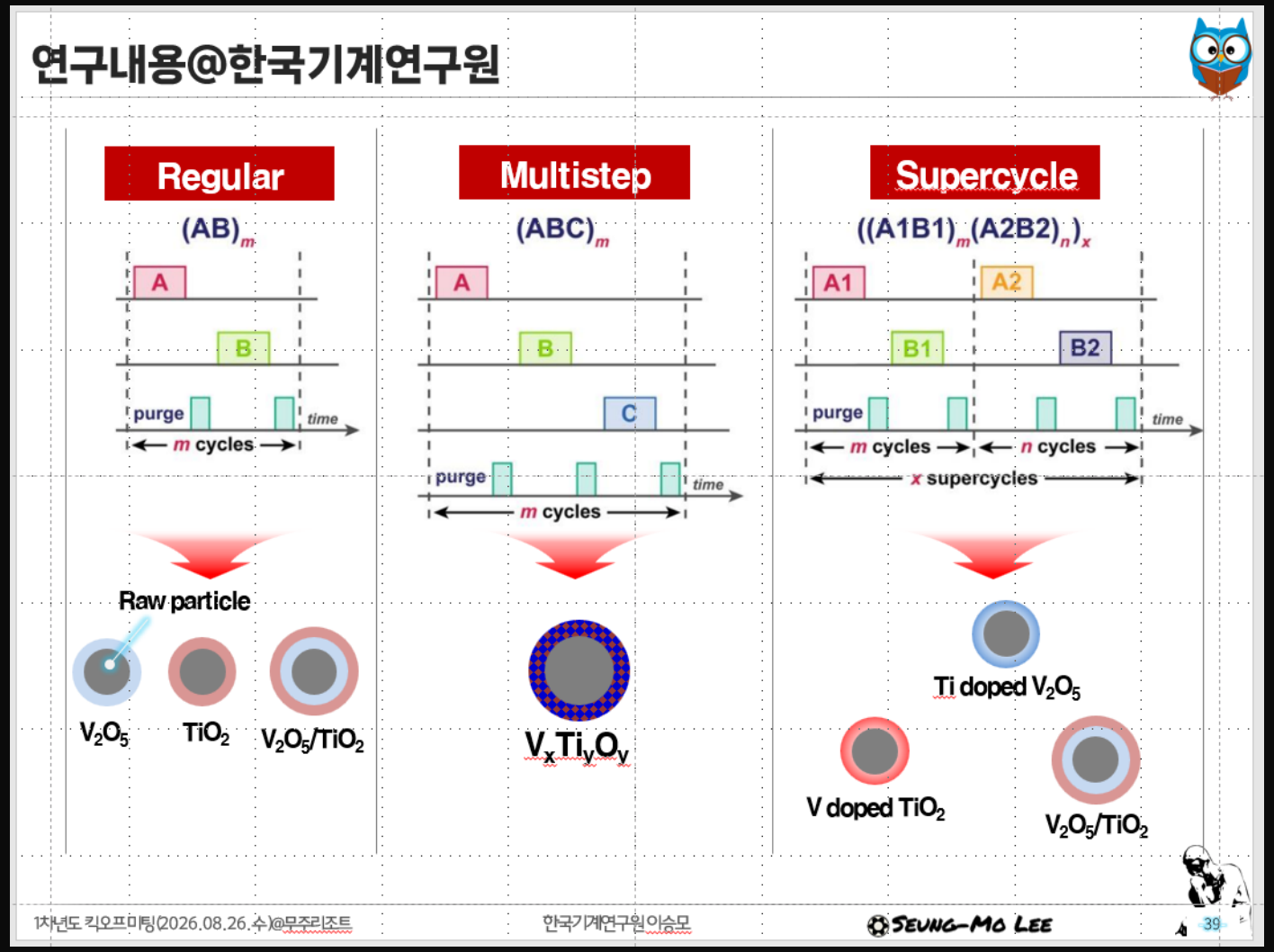
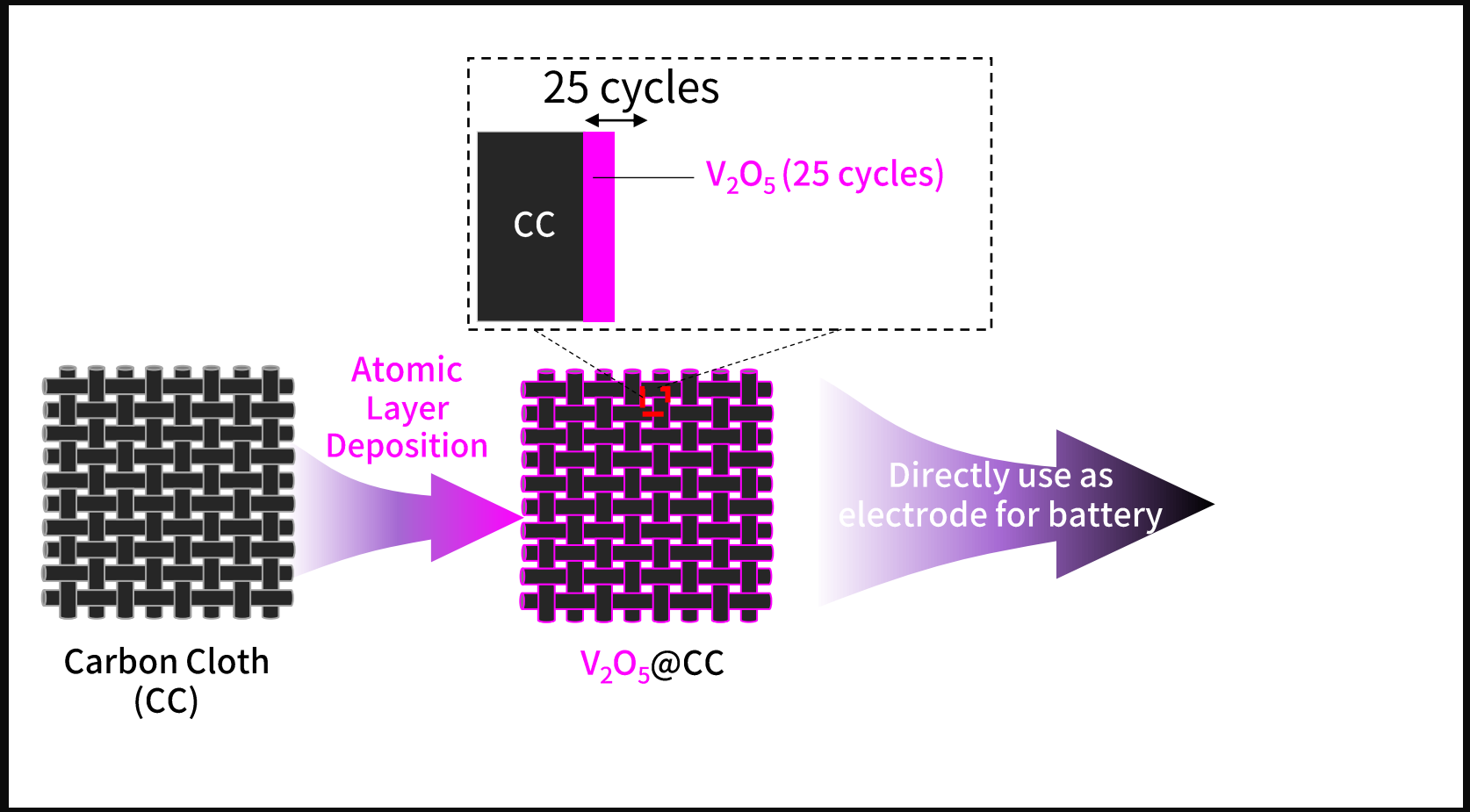
TiO_2–V_2O_5 Heterointerfaces on Carbon Cloth for Enhanced Li-ion Storage

Uyen NTN
Korea University of Science and Technology (UST)

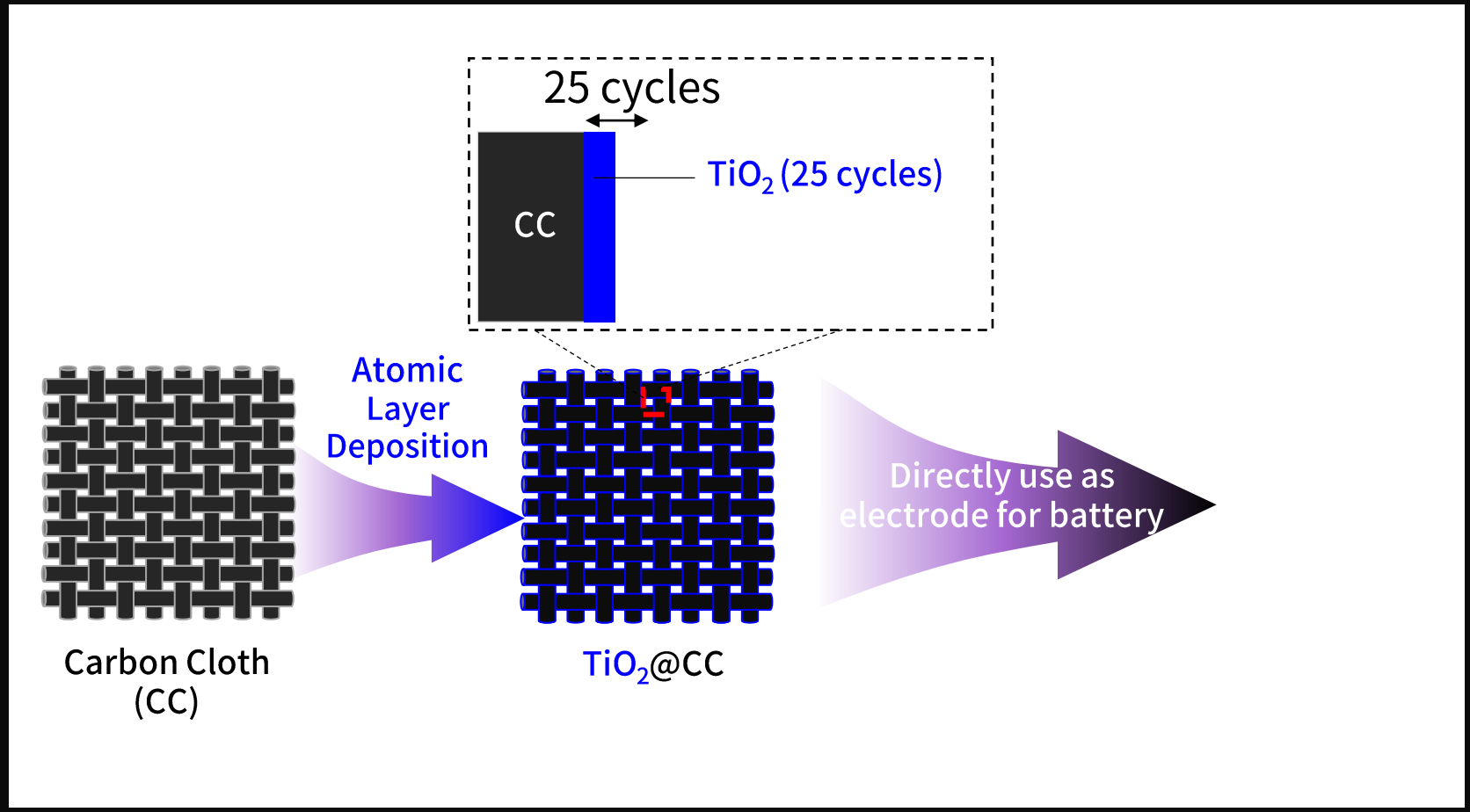
Our approach



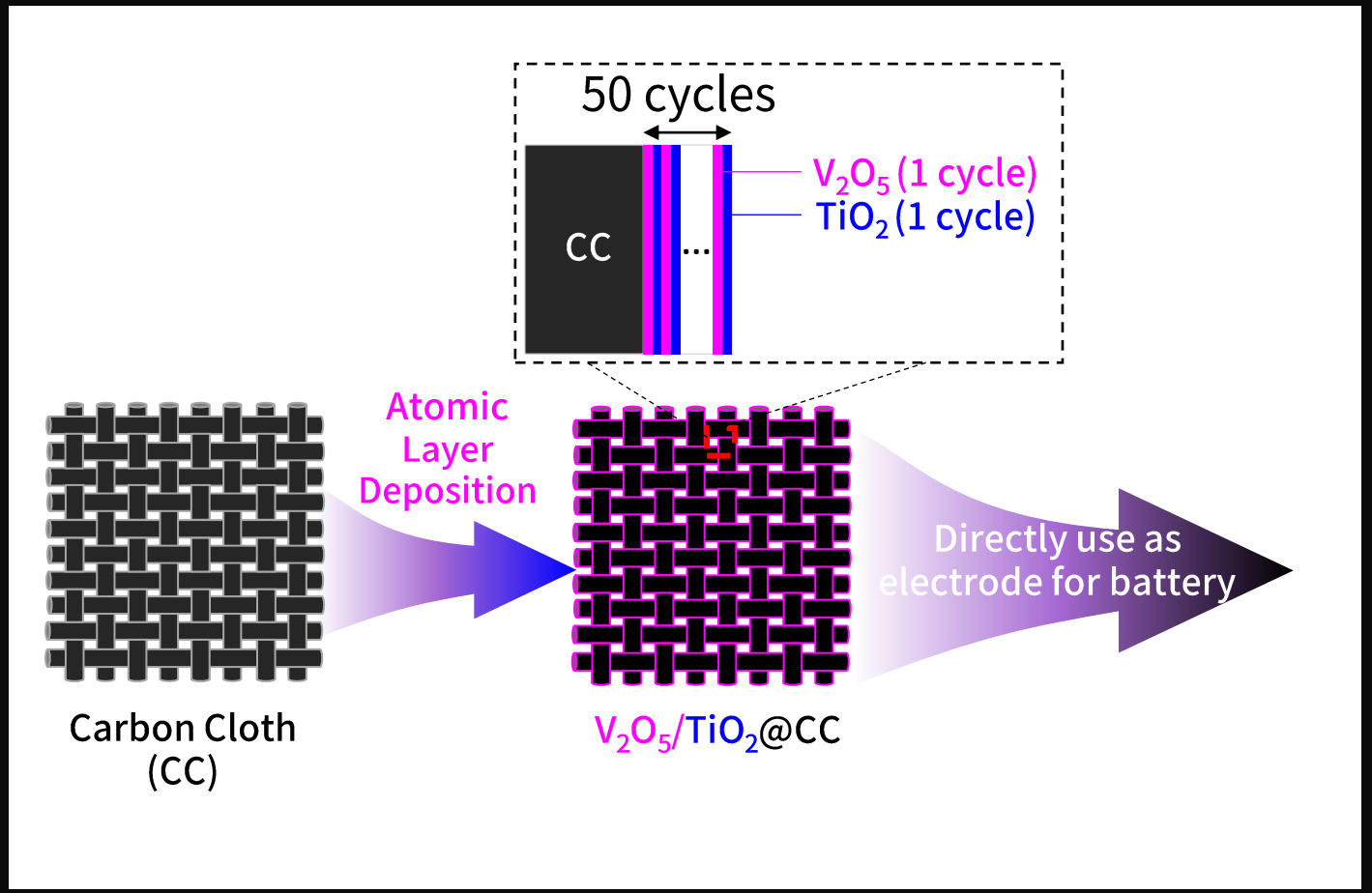
Regular sample



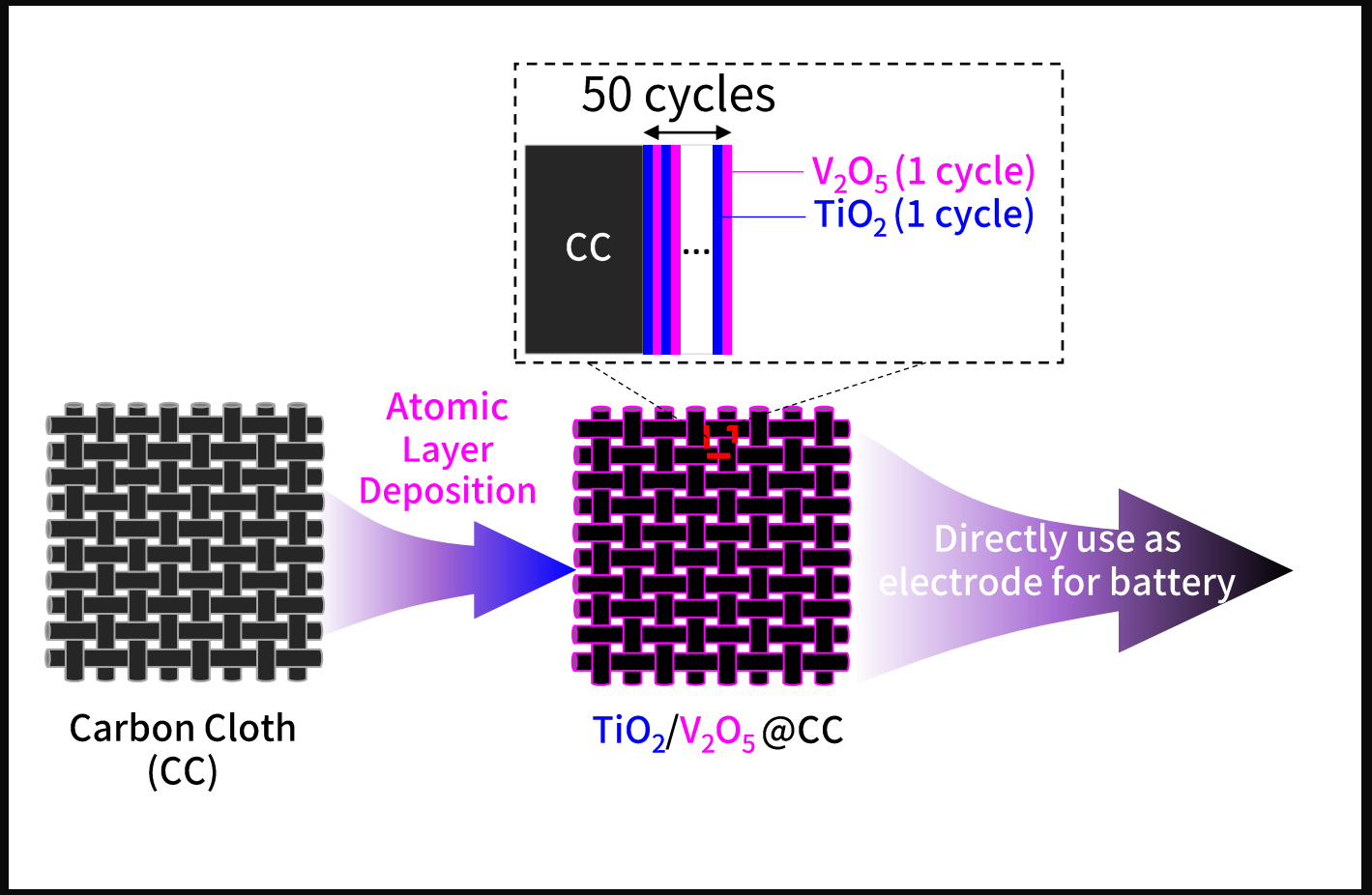
Regular sample



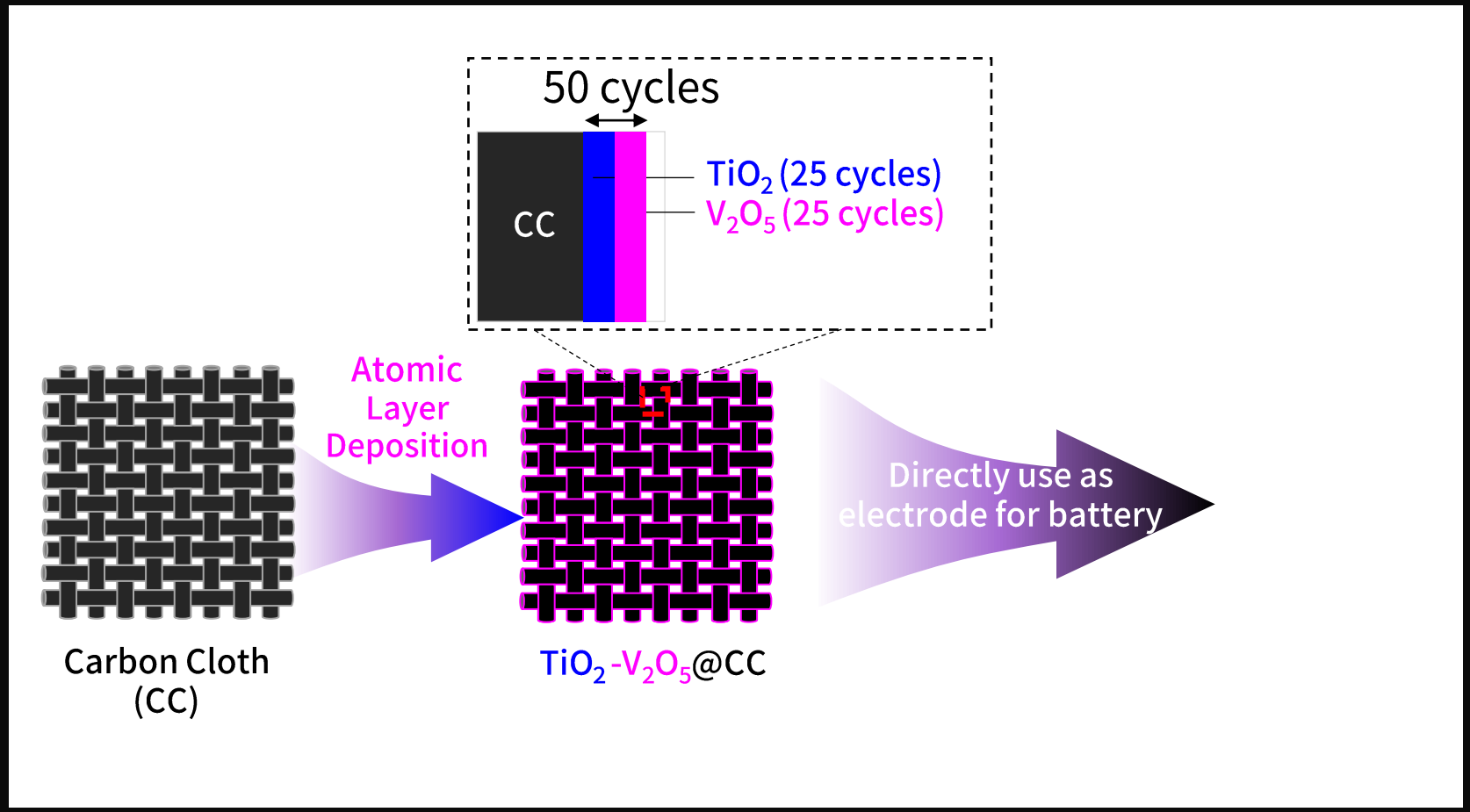
Supercycle sample



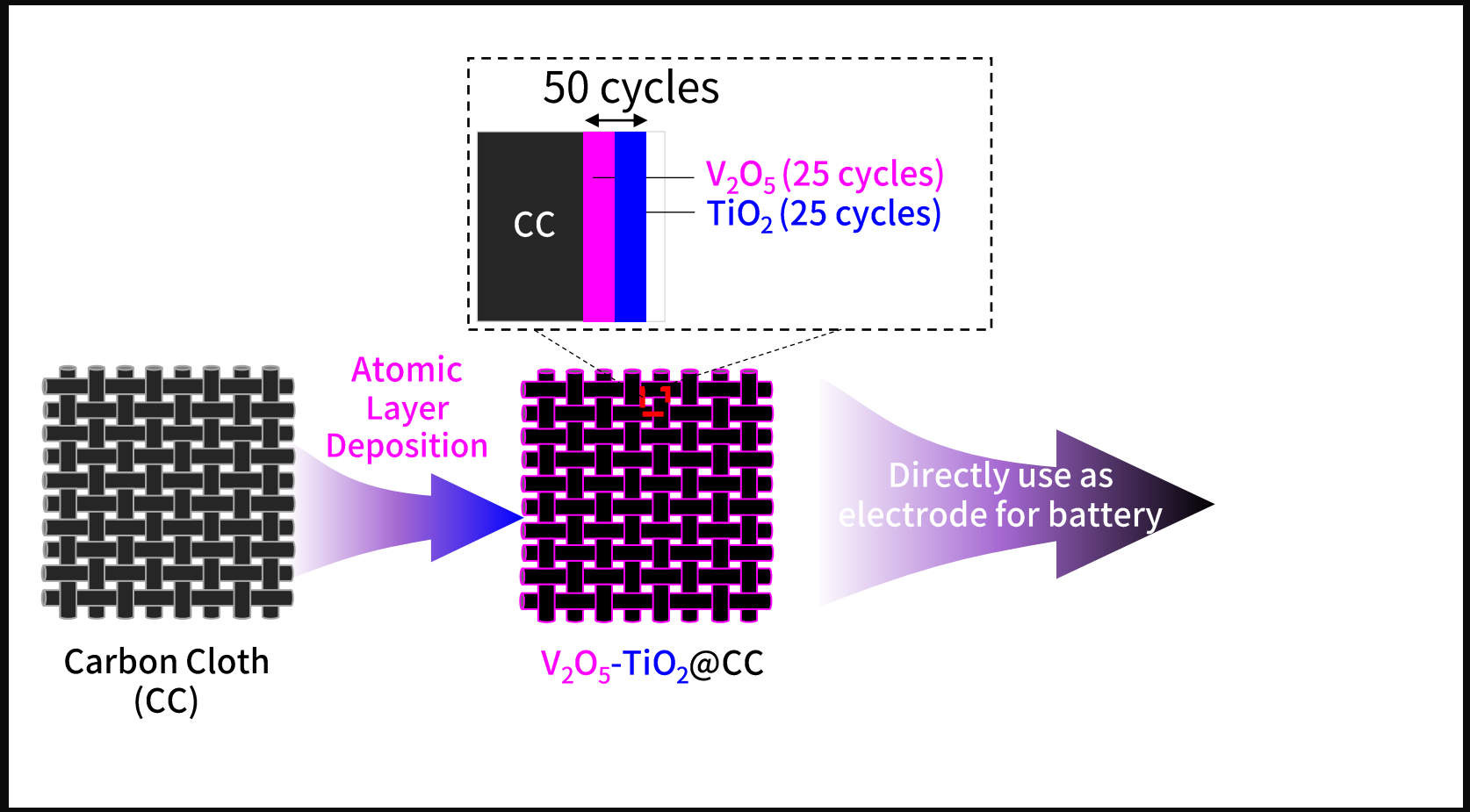
Supercycle sample



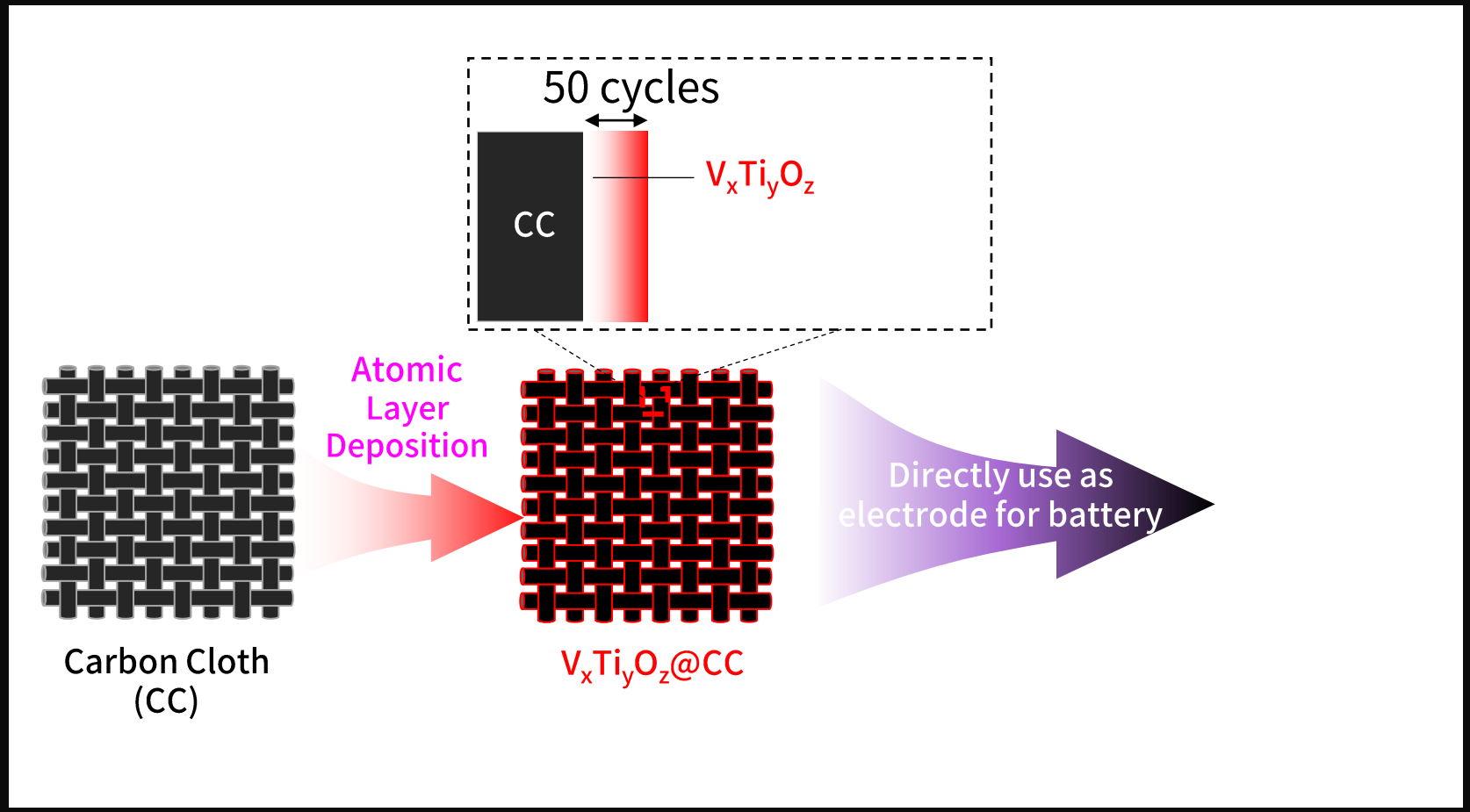
Supercycle sample



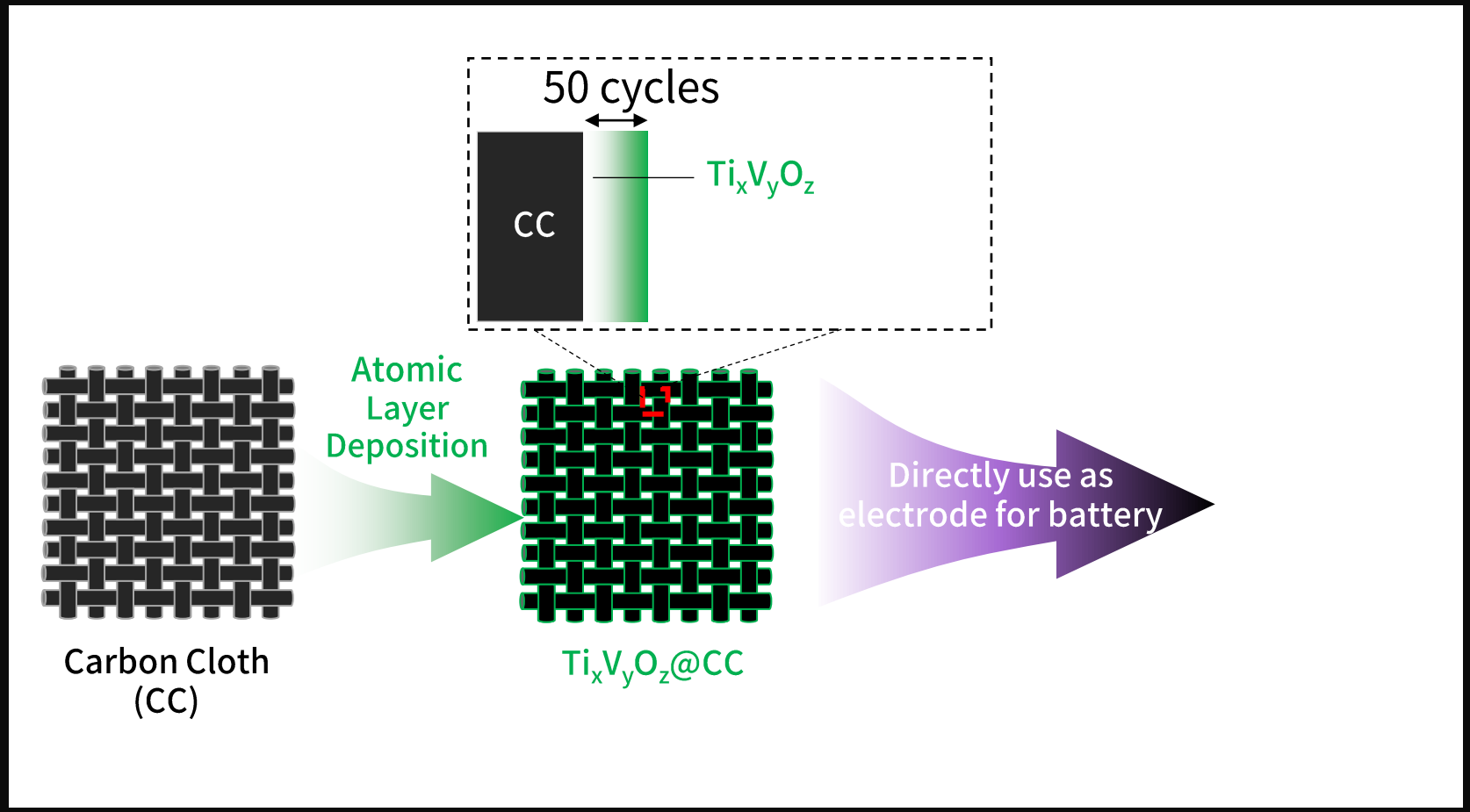
Supercycle sample



Multistep sample



Multistep sample

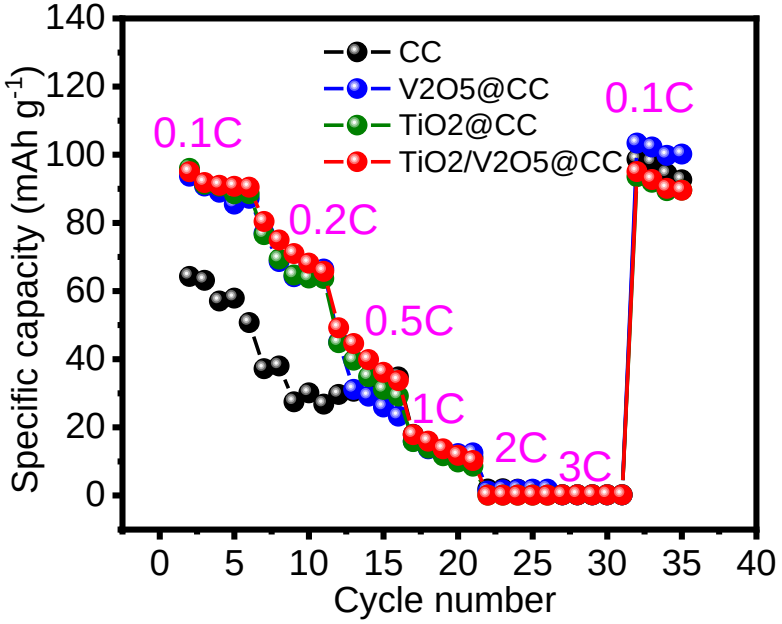


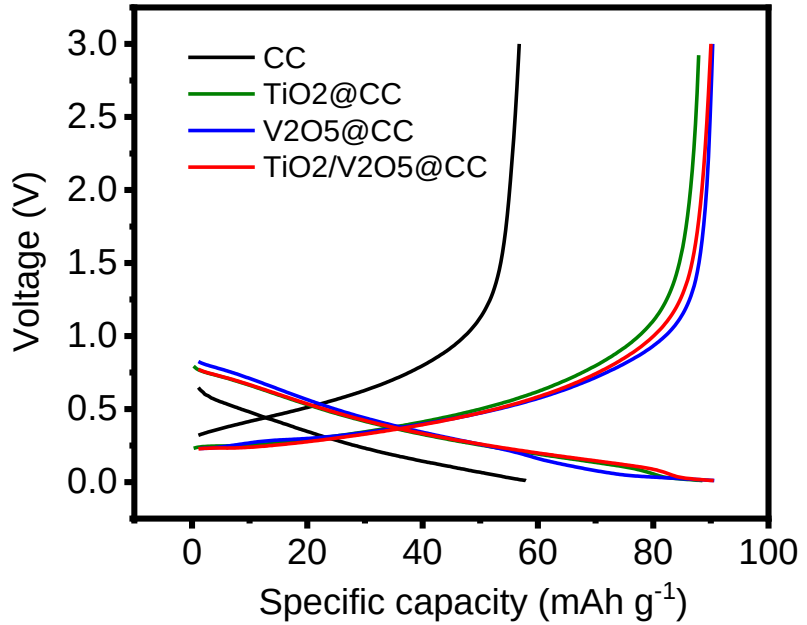
Our approach

Number of cycles / Sample	ALD	Li ion measurement
CC	-	O
V_2O_5@CC	O	O
TiO_2@CC	O	O
V_2O_5/TiO_2@CC	O	O
TiO_2/V_2O_5@CC	O	O
TiO_2 -V_2O_5@CC	O	O
V_2O_5-TiO_2@CC	O	O
$V_xTi_yO_z$@CC	Not yet	Not yet
$Ti_xV_yO_z$@CC	Not yet	Not yet

First test (Li ion Battery)

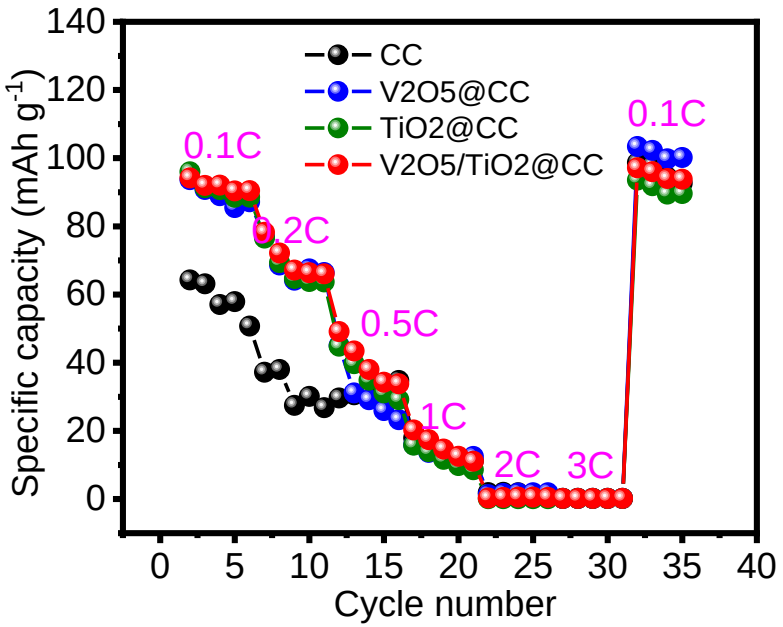
2026.09.03

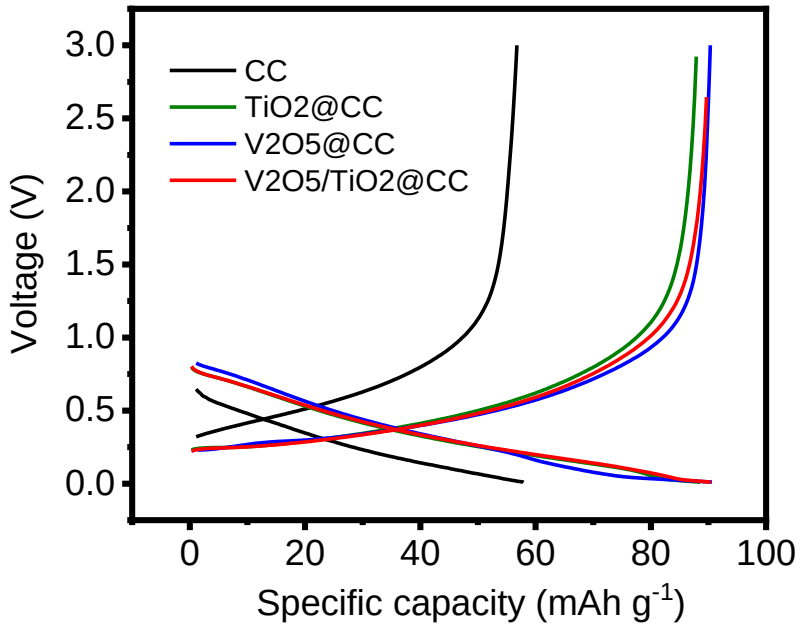




First test (Li ion Battery)

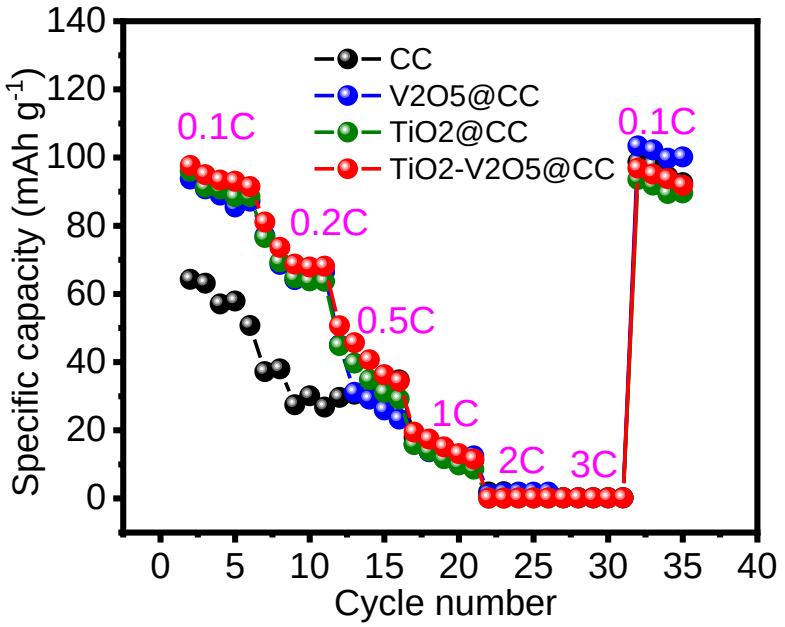
2026.09.07

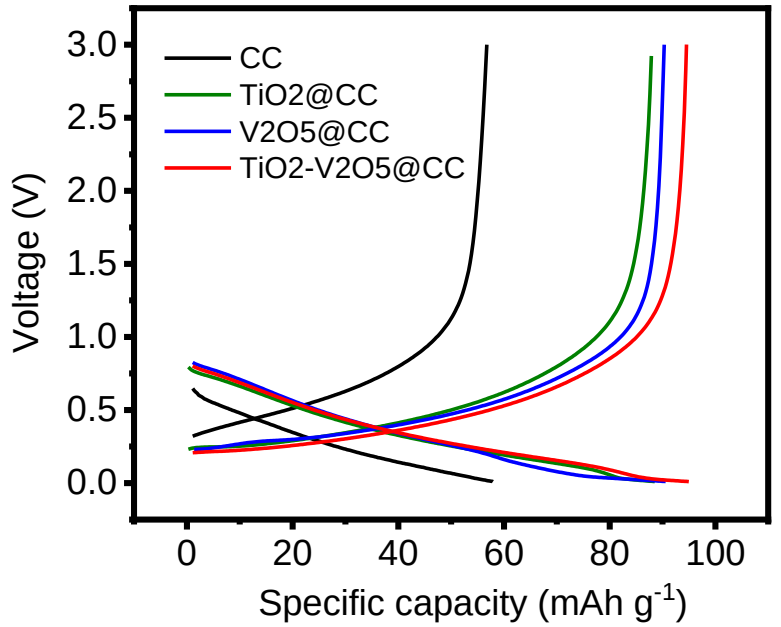




First test (Li ion Battery)

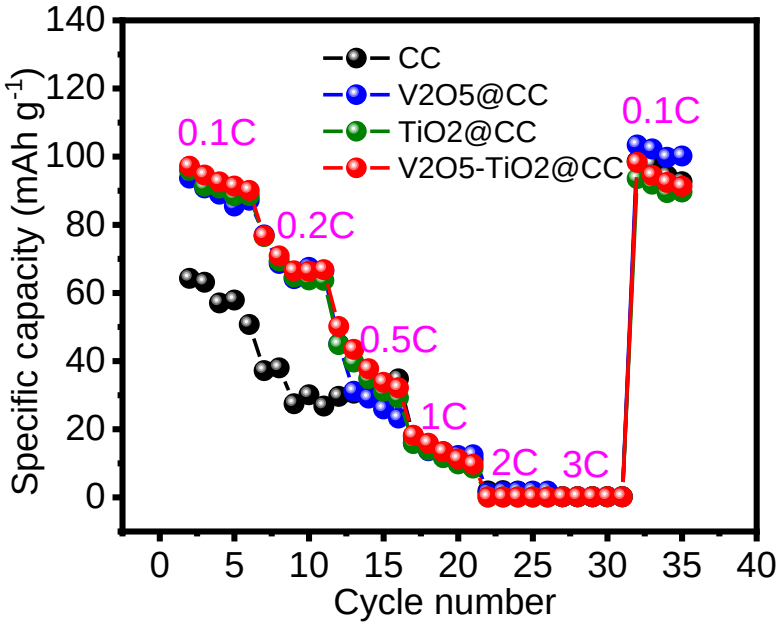
2026.09.07

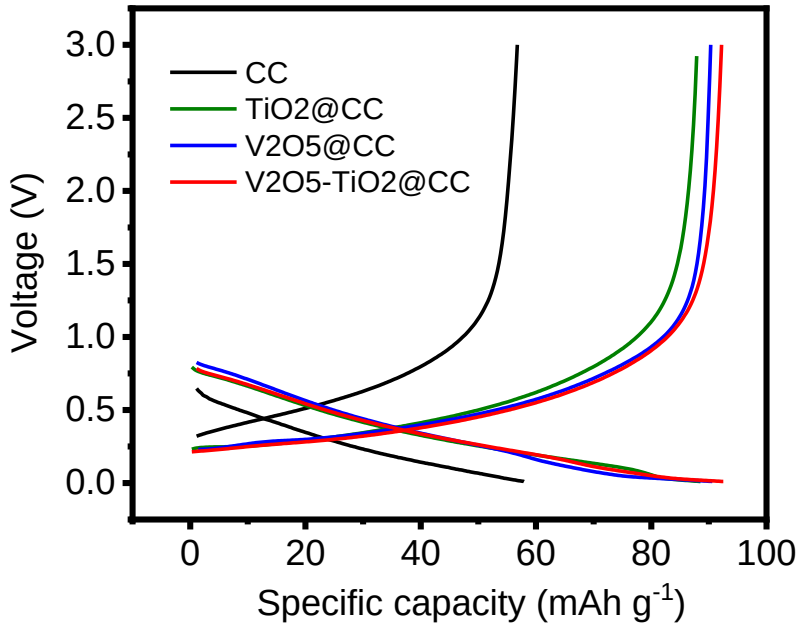




First test (Li ion Battery)

2026.09.07





Temporary Conclusion

Multistep TiO_2–V_2O_5@CC and V_2O_5–TiO_2@CC show slightly better performance than the corresponding supercycle samples, compared with CC, V_2O_5@CC, and TiO_2@CC.

→ Further test additional multistep ALD configurations.